

Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 3



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Recap of Previous Lecture



Topic

Topic

Topic

Regular languages

DFA NFA RE

Topics to be Covered



Topic

Topic

Topic

Regular languages

$\Sigma = \{a, b\}$ $L = \{w \mid |w| = \text{even}\} \rightarrow \text{even length strings}$

$L = \{\epsilon, aa, ab, ba, bb, aaaa, bbbb, abab, \dots\}$

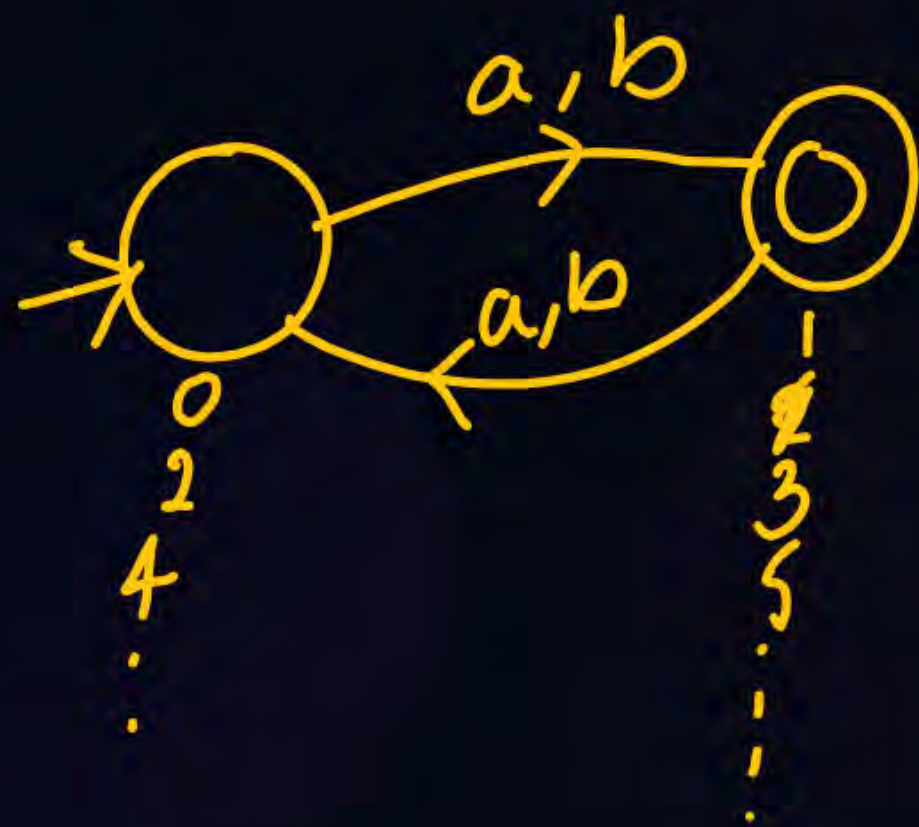
$$(a+b)(a+b)^* = (\underline{a}b + b\underline{a} + b\underline{b} + \underline{a}a)^*$$

ababab



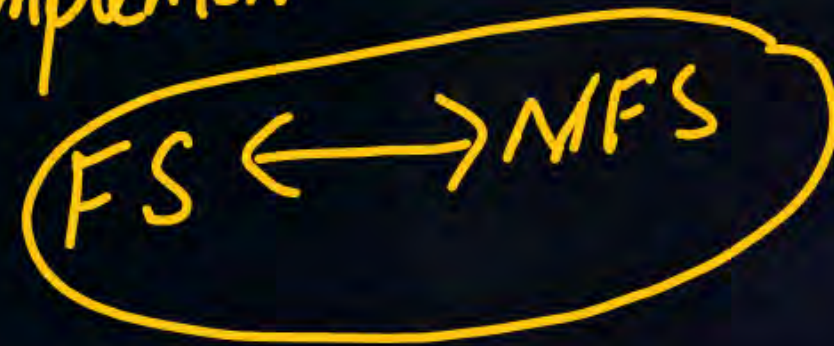
$\Sigma = \{a, b\}$ $L = \{w / |w| \text{ is odd}\}$
 $L = \{\underline{a}, \underline{b}, a\underline{a}, a\underline{a}b, b\underline{b}b, \dots\}$

$$(a+b)((a+b)(a+b))^*$$



$$((a+b)(a+b))^* \underline{(a+b)}$$

Complement



$$\begin{array}{c} (a+b)^* \\ \downarrow \\ U \\ X \end{array}$$

$$RE \rightarrow (a+b)^{2n+1} / n \geq 0 \quad \checkmark$$

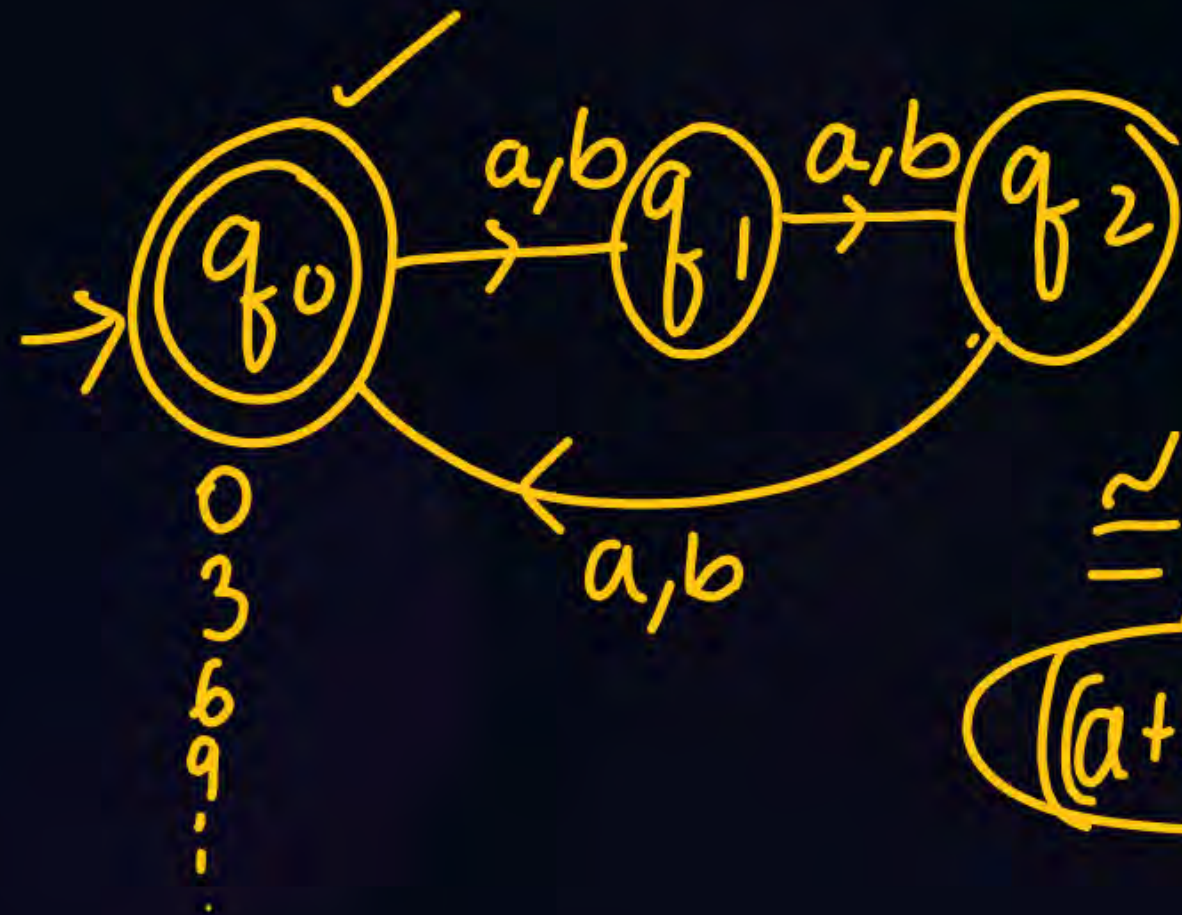
$$\underline{(a+b)^{2n}} \underline{(a+b)} / n \geq 0$$

$$\oplus \checkmark$$

$$(a+b)(a+b)^*(a+b)$$

$$a^+ \quad a^{*+}$$

$\Sigma = \{a, b\}$ $L = \{w / |w| \equiv \underline{\underline{0 \pmod{3}}}\}$



$$(a+b)(a+b)(a+b)^*$$

$$\not\equiv 0 \pmod{3}$$

$$(a+b)(a+b)(a+b)(a+b)(a+b)^*$$

$$(a+b)(a+b)(a+b)^*$$

$$\begin{aligned} &\equiv 1 \pmod{3} \rightarrow q_1 \checkmark \equiv 2 \pmod{3} \downarrow q_2 \\ &\underline{\underline{(a+b)(a+b)(a+b)(a+b)^*}} \end{aligned}$$

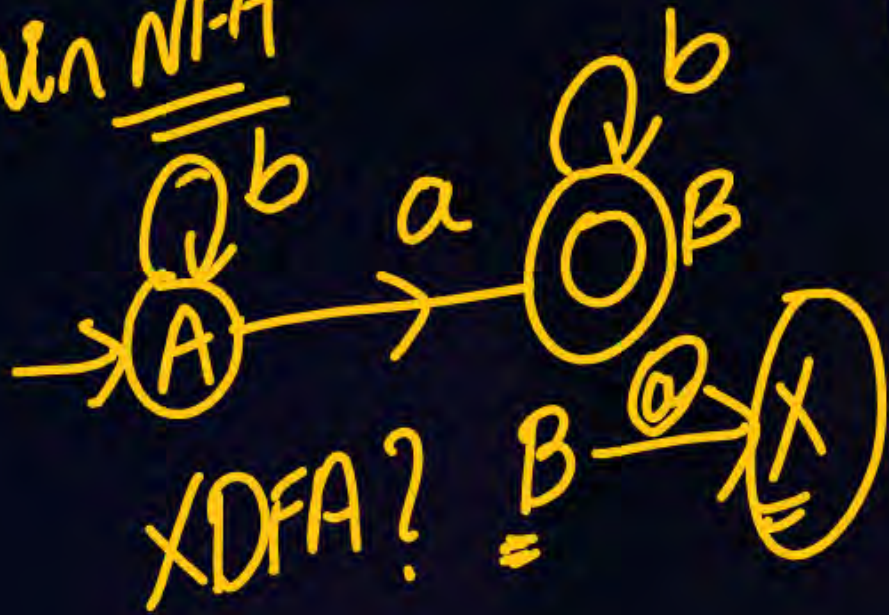
$\Sigma = \{a, b\}$

$L = \{w / w \text{ contains exactly one } \underline{a}\}$

RE: $b^* a b^*$ ✓

$L = \{ \underline{a}, \underline{b}a, a\underline{b}, \underline{b}ba, bab, \dots \}$

min NFA



min DFA



aabbX

abaX

ababX

abb✓

minimization algo

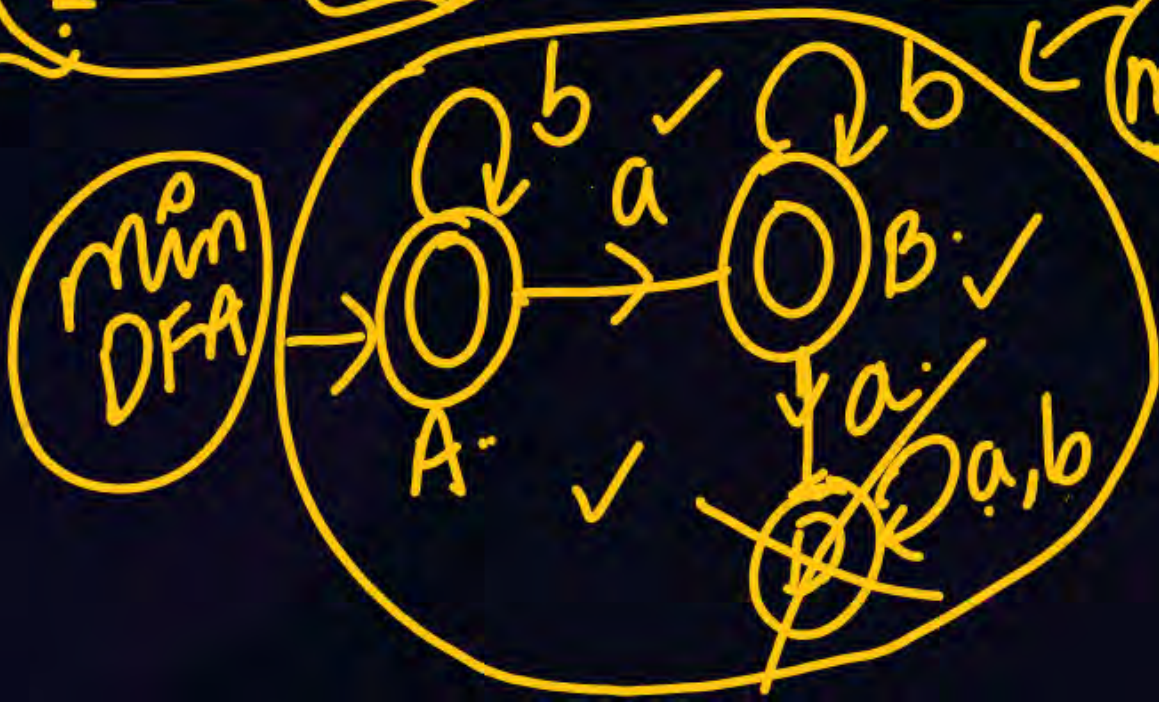
$\Sigma = \{a, b\}$

$L = \{w/w \text{ contains at most } 1a\} \rightarrow \underline{0a's} + \underline{1a's}$

~~$\epsilon + b^* + b^*ab^*$~~ = $b^*(\epsilon + a)b^*$

$\epsilon + aa$

$L = \{\epsilon, aa\}$



min NFA

2 states min DFA - 3 states

$$\Sigma = \{a, b\}$$

$$L = \{w/w \text{ contains}\}$$

$$\text{No of states in min } \underline{\text{NFA}} \leq \text{no of states in min } \underline{\text{DFA}}$$

$$\underline{N+1} \quad \underline{O(N)} \checkmark \quad \leq \quad \underline{2^N} \quad \underline{O(2^N)} \checkmark$$

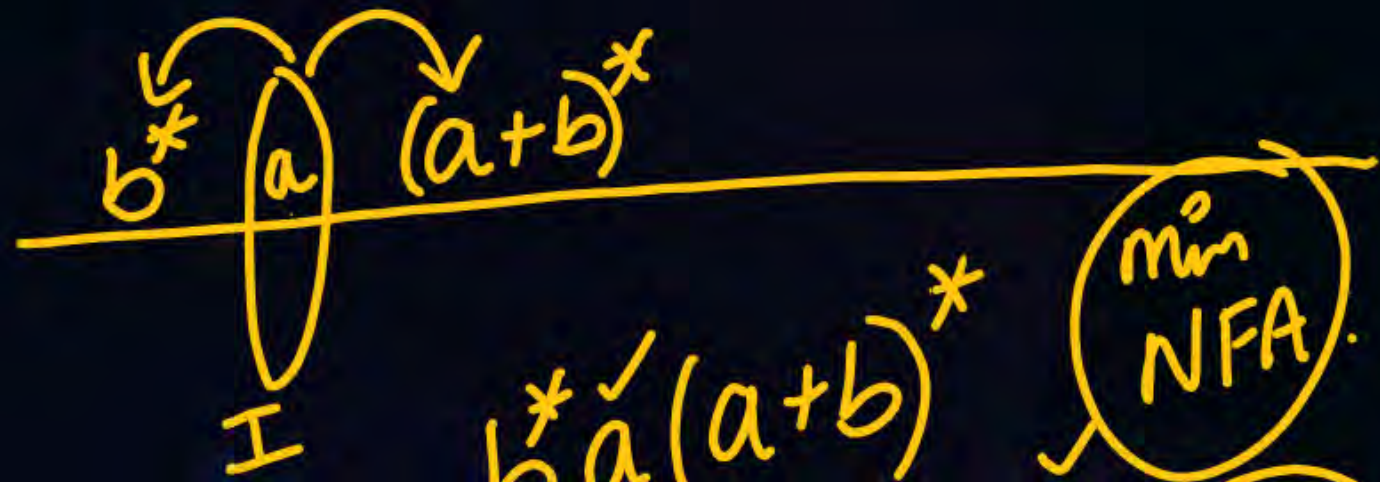
min NFA will never have dead state

$\Sigma = \{a, b\}$

$L = \{w/w \text{ contains at least } 1a\}$

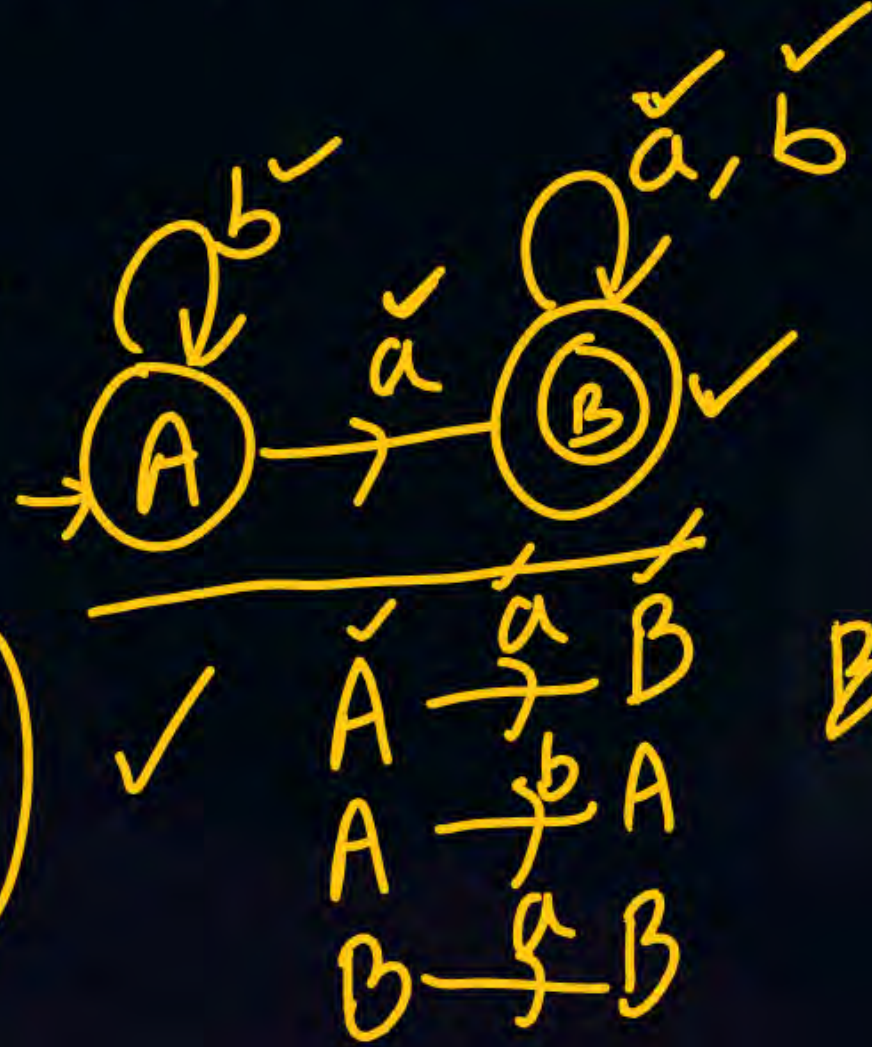
Exactly 1a → Dead
Atmost 1a → Dead
Atleast 1a → No Dead

DFA



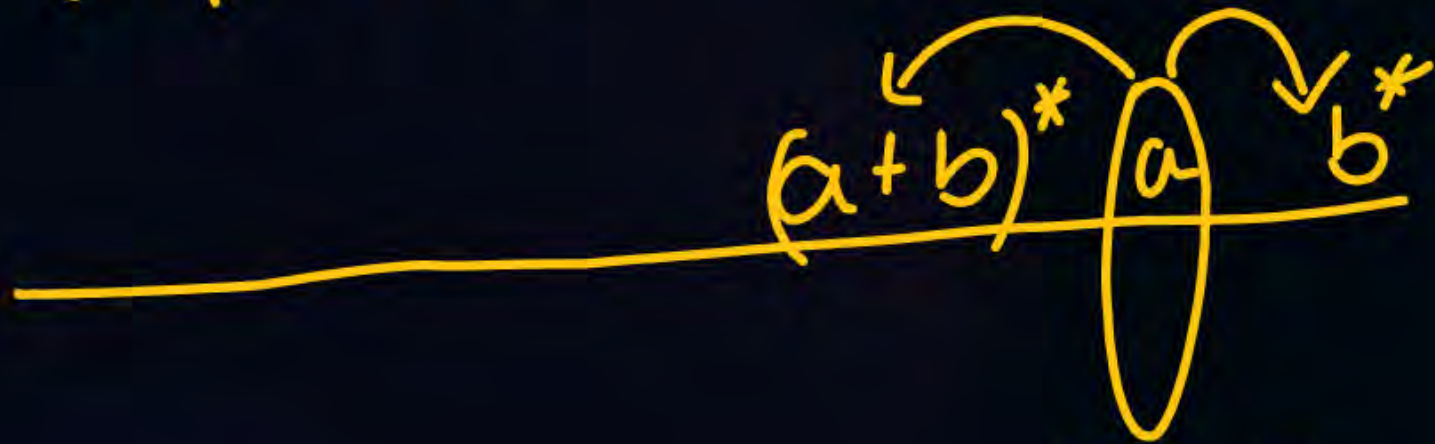
RE $b^* a (a+b)^*$

min NFA
min DFA



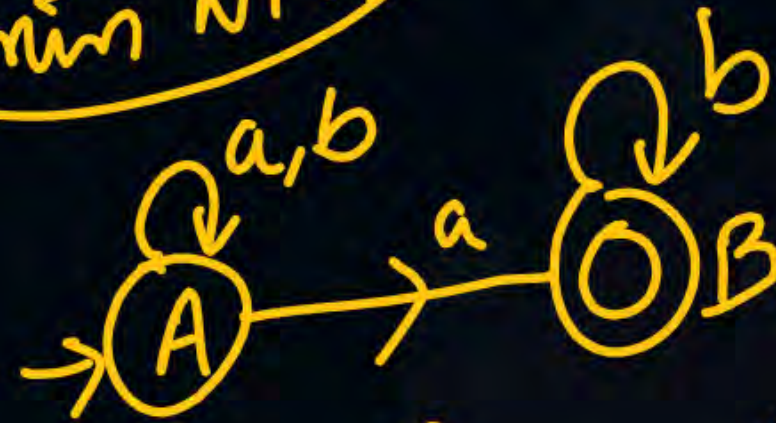
$\Sigma = \{a, b\}$

$L = \{w/w \text{ contains at least } 1a\}$

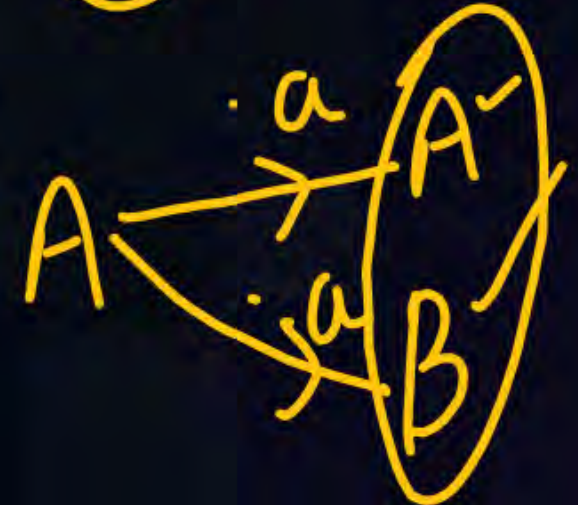


last a $\neq$
 $(a+b)ab^*$

min NFA ✓



DFA?
 $\times$

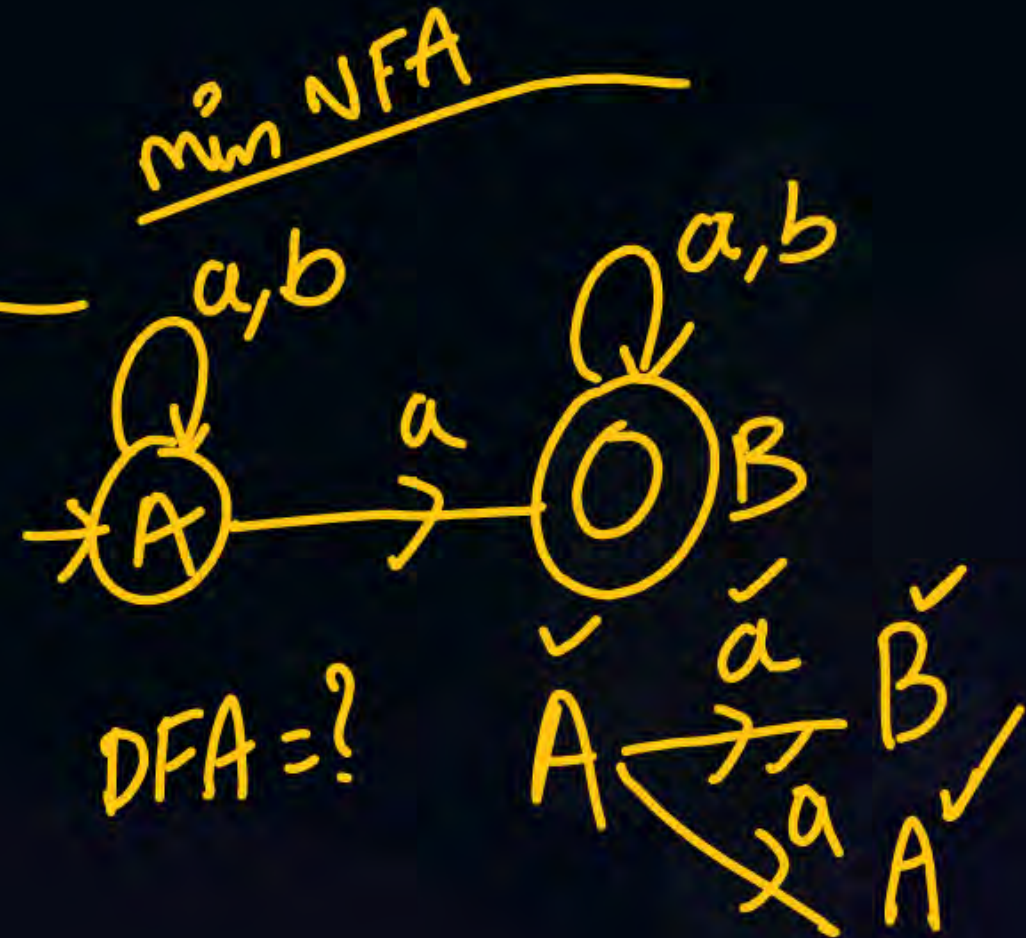


$\Sigma = \{a, b\}$

$L = \{w/w \text{ contains at least } 1a\}$

$(a+b)^* a (a+b)^*$

Some random a
 $(a+b)^* a (a+b)^*$



10 minutes

exactly 2 a's

atleast 2 a's

atmost 2 a's

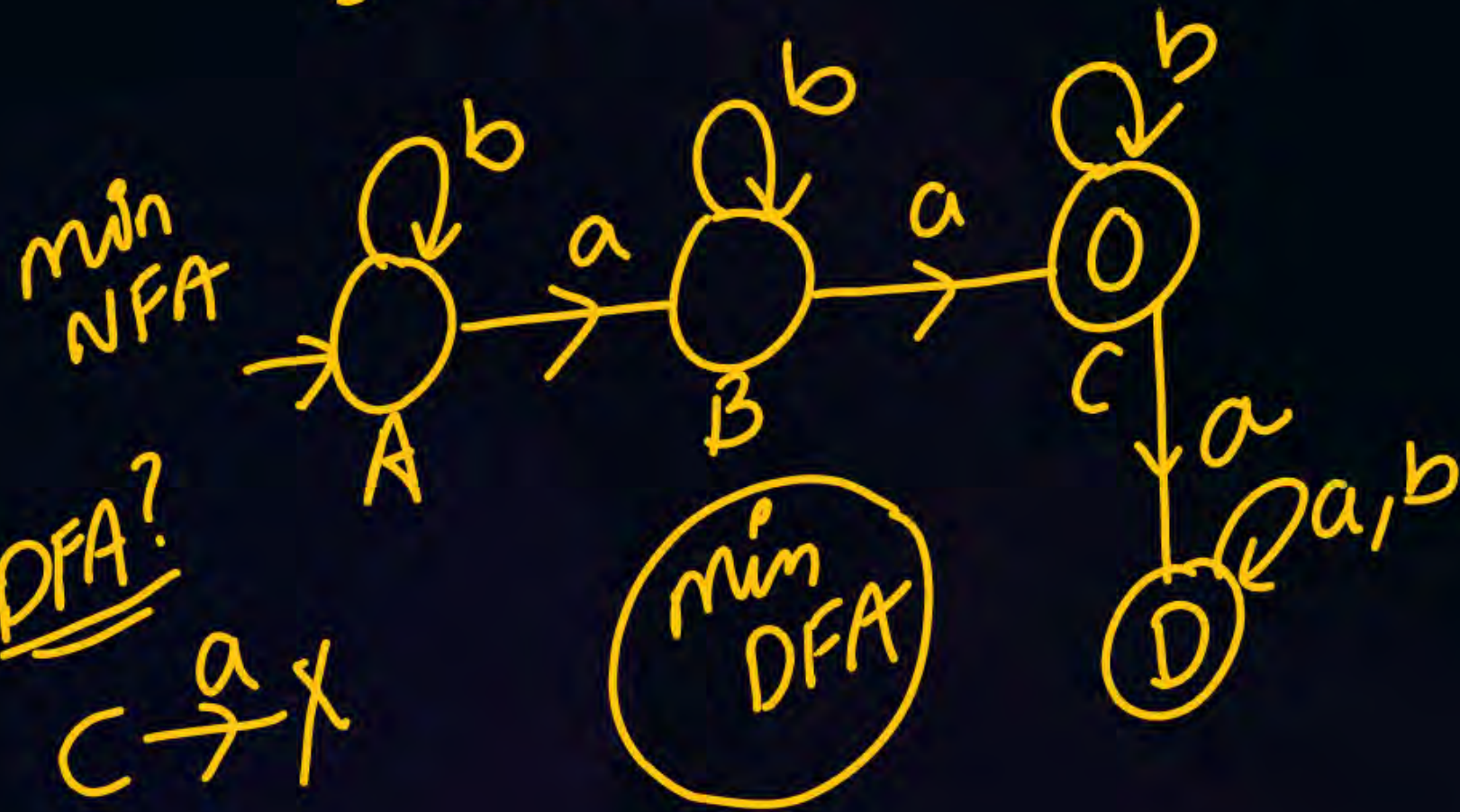
min DFA

min NFA

min REG X

exactly 2 a's
 $\underline{b^*} \underline{a} \underline{b^*} \underline{a} \underline{b^*}$

$b^* \overset{+}{\underline{a}} a b^* \times$



at most 2 a's.

$$\cancel{A} + \underline{b^*} + b^*ab^* + b^*ab^*ab^*$$

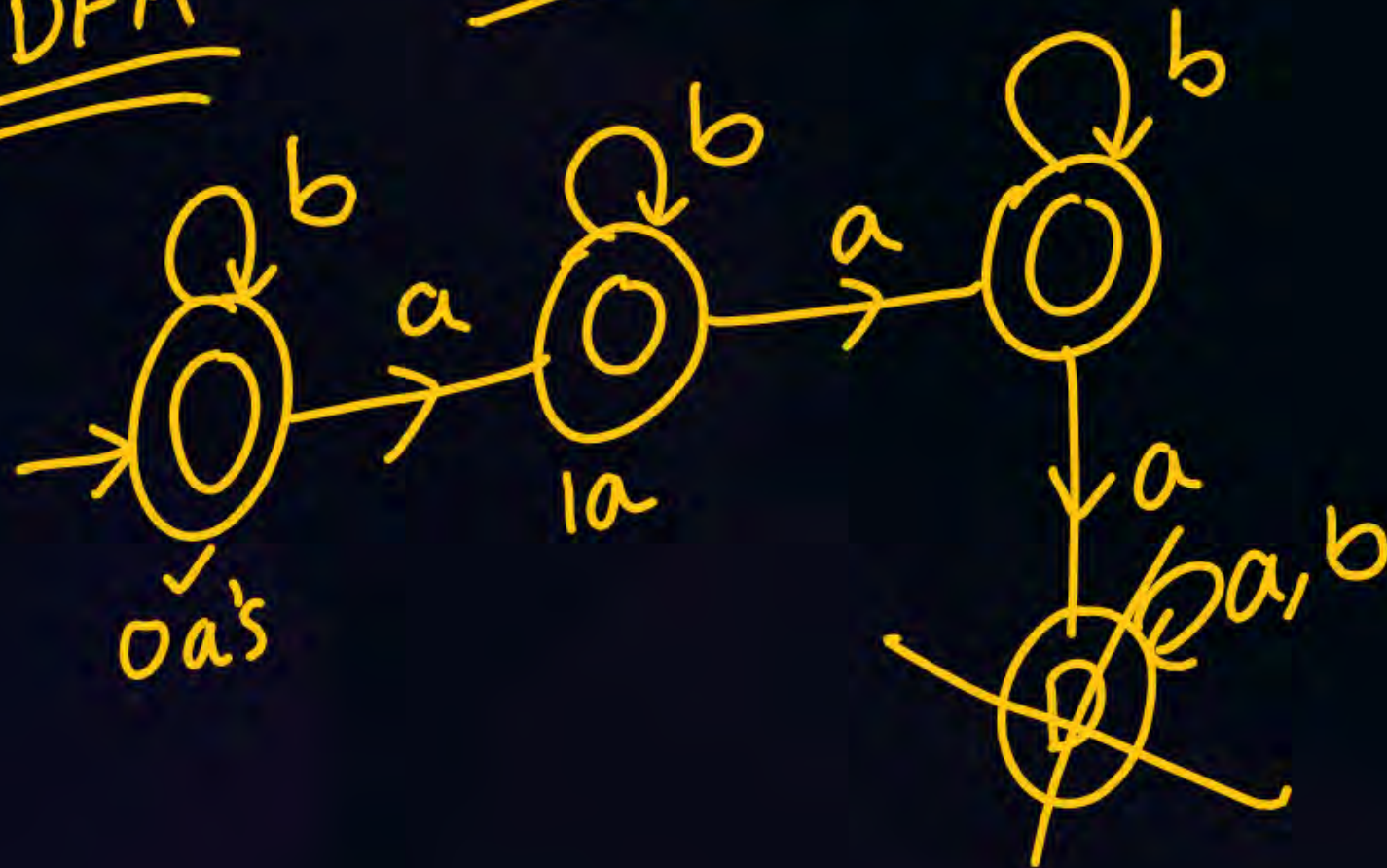
$$b^*(\underline{a+e})b^*(\underline{a+e})b^*$$

$$\underline{(a+b)^*} \underline{a} \underline{(a+b)^*} \underline{a} \underline{(a+b)^*}$$

at most 2 a's

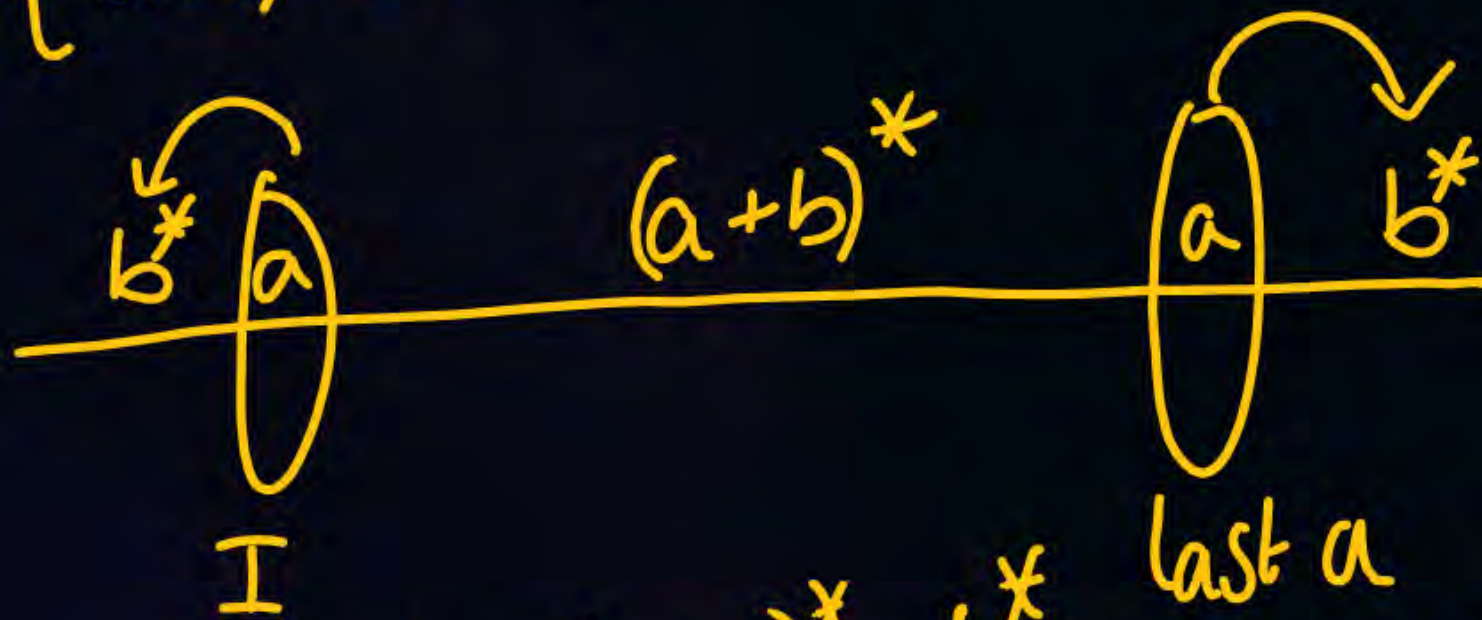
min DFA

min NFA ✓



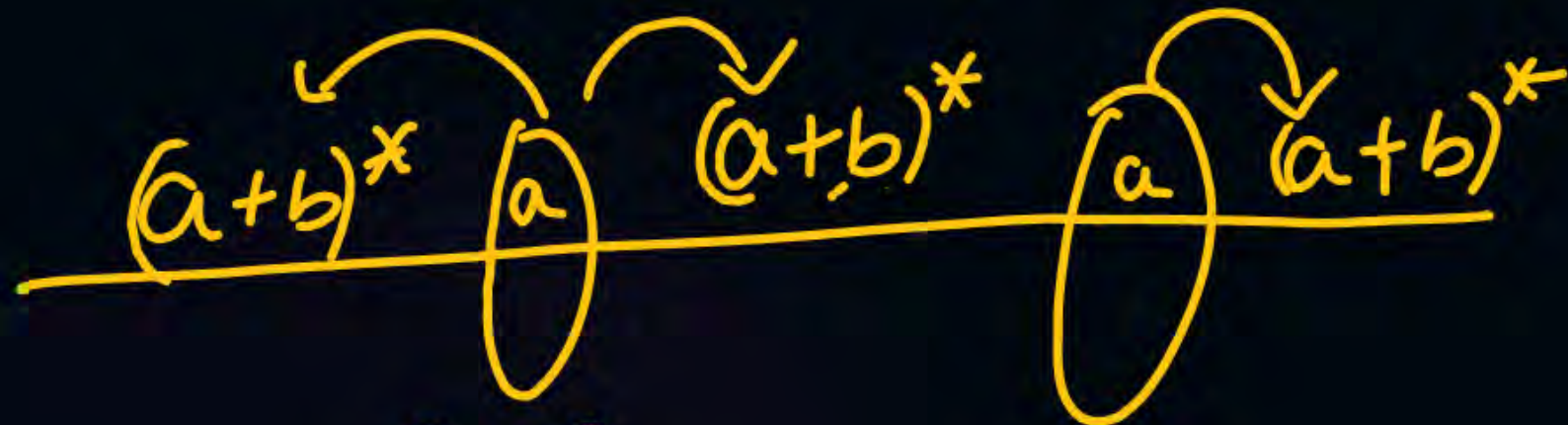
atleast 2 a's

$L = \{aa, baa, aba, aab, aaa, baaa, \dots\}$



$b^* a (a+b)^* a b^*$





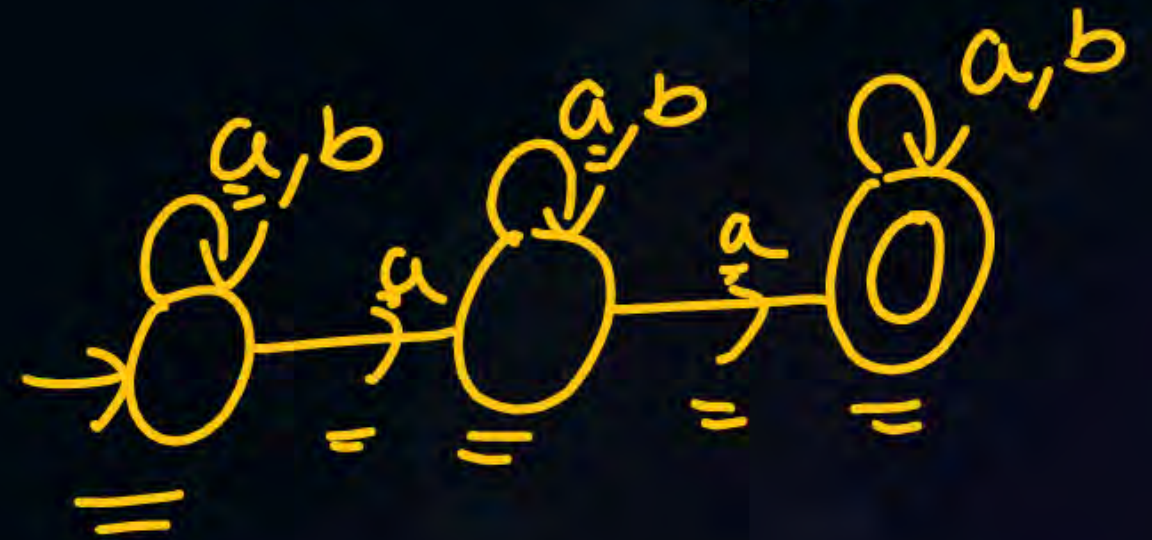
Some
random
 a

Some
random
 a

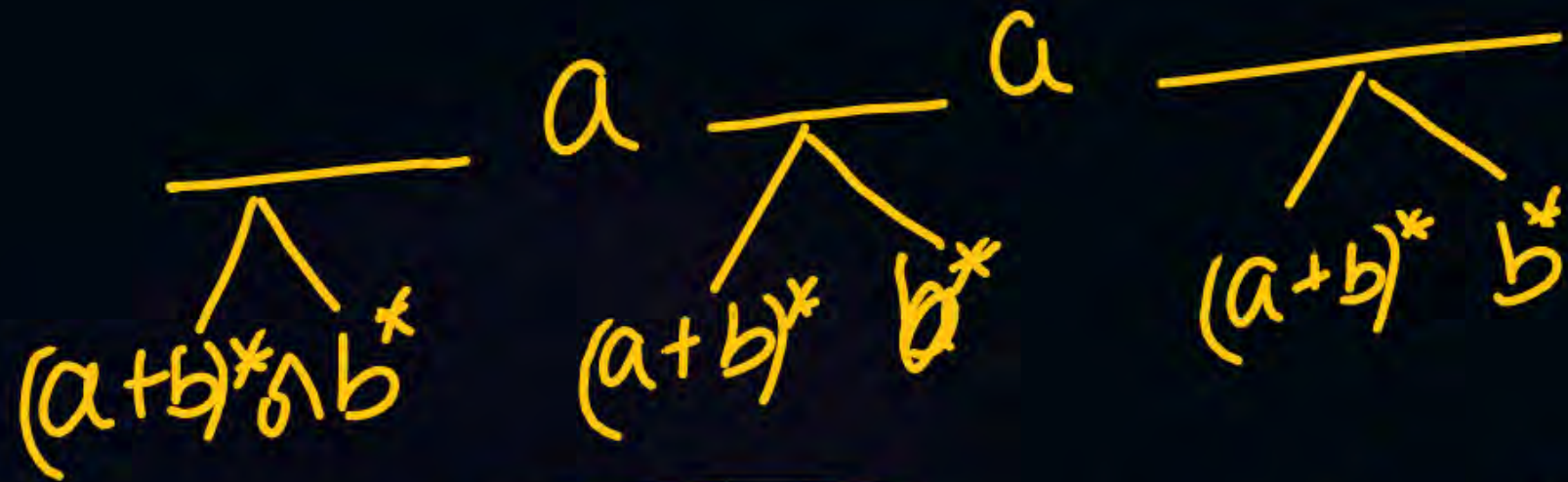
$(a+b)^* a (a+b)^* a (a+b)^*$

min NFA

$\times$



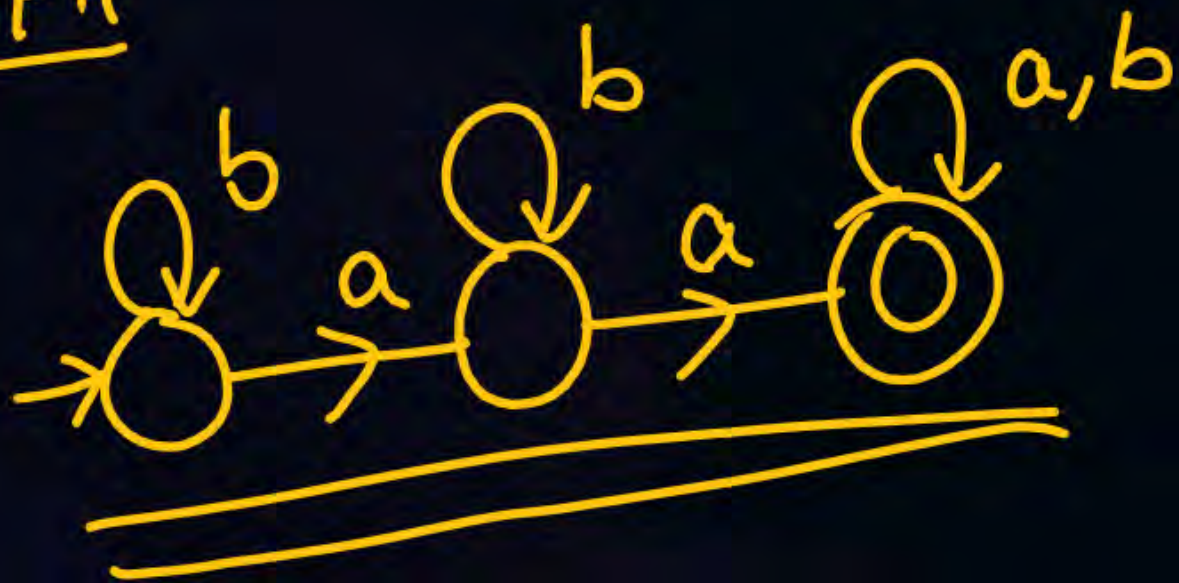
$$\overline{2} \times \overline{2} \times \overline{2} = \underline{\underline{(8)}} - \underline{\underline{1}} = \underline{\underline{(7)}}$$



$b^*ab^*ab^*$ → exactly 2 a's
 7 RE → 7 min NFA

atleast 2 a's

min DFA



$$(a+b)^* = (a+b^*)^* = (a^*+b^*)^* = (a^*+b)^* = (a^*b^*)^* = (b^*a^*)^*$$

a, b

(a+b)*

$\Sigma = \{a, b\}$ even no of a's

~~$$\emptyset + b^* + (b^*ab^*ab^*)^*$$~~

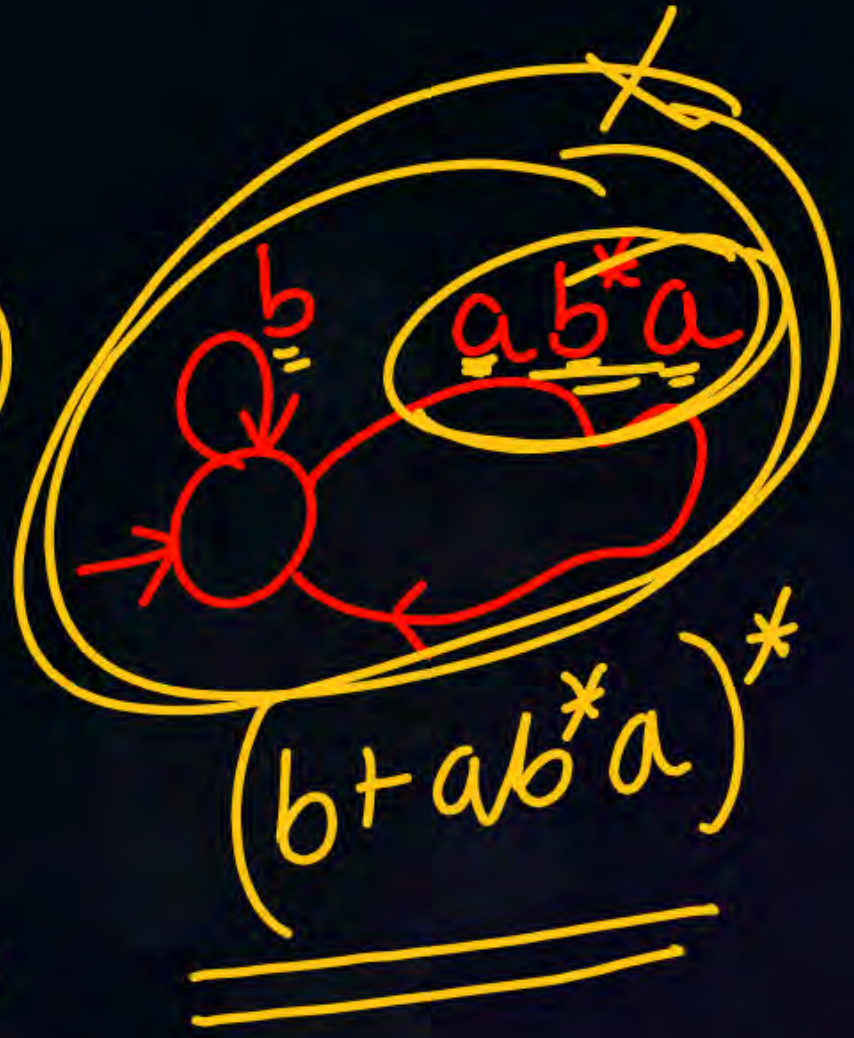
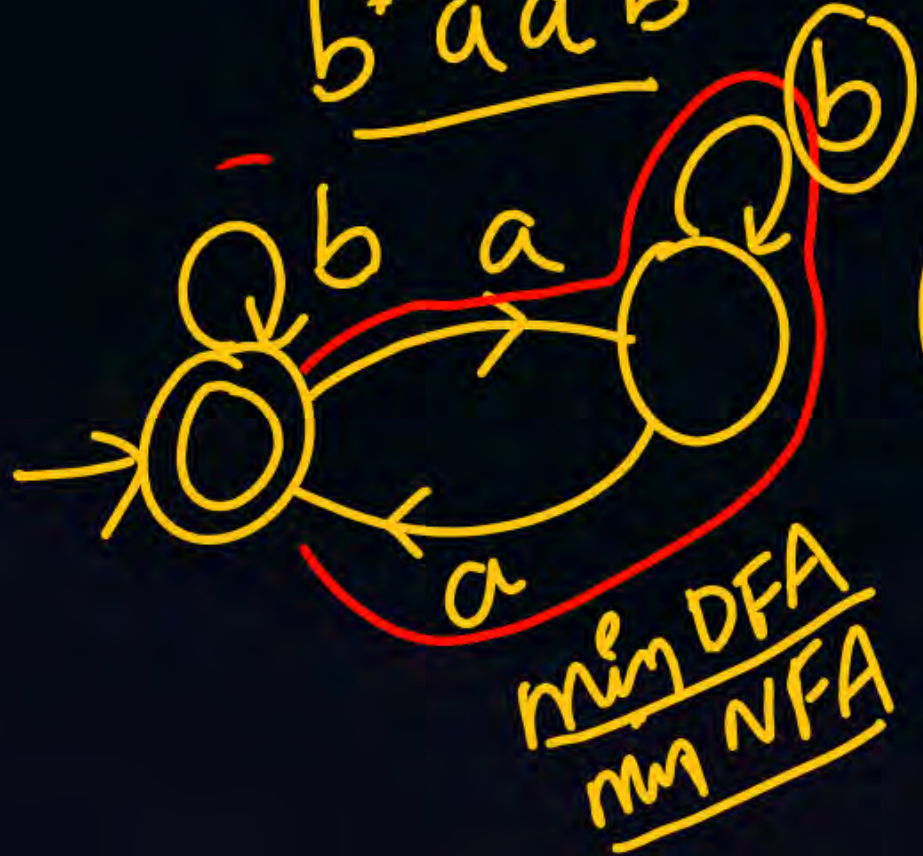
$$(b+ab^*a)^*$$

$$(b^*(ab^*a)^*)^*$$

$$b^*aab^*$$

$$b^*ab^*ab^*$$

$$b^*aab^*$$

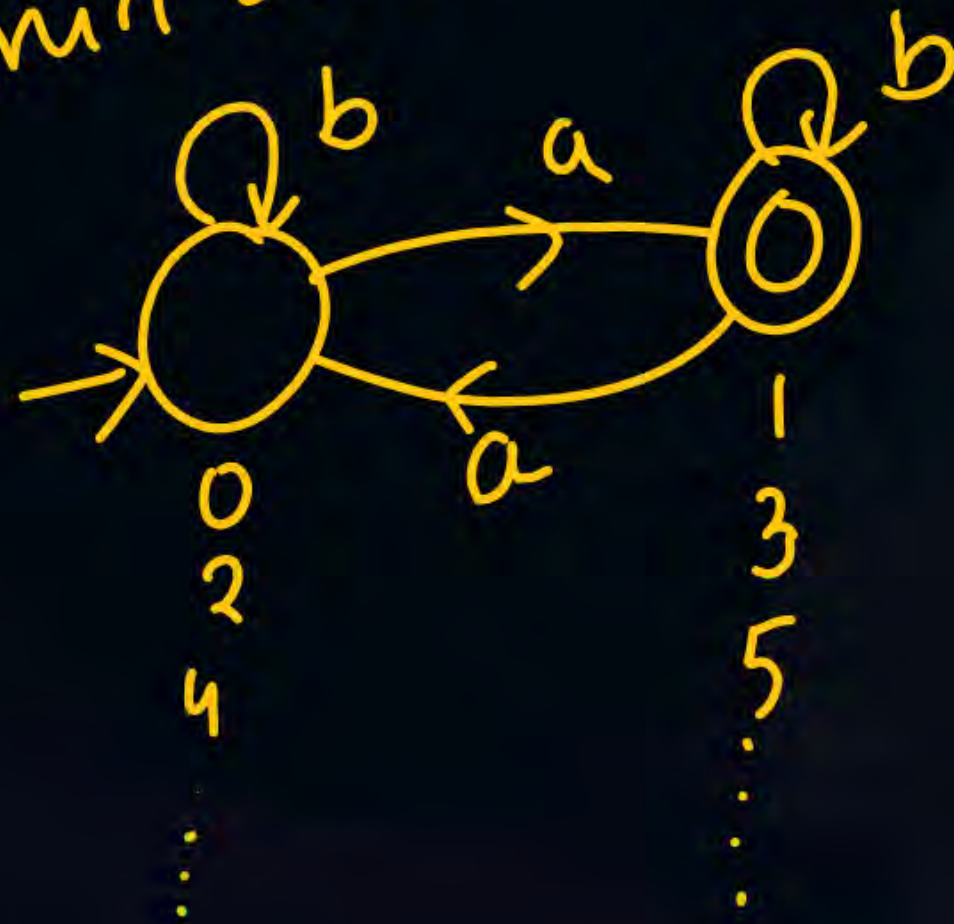


$$\Sigma = \{a, b\}$$

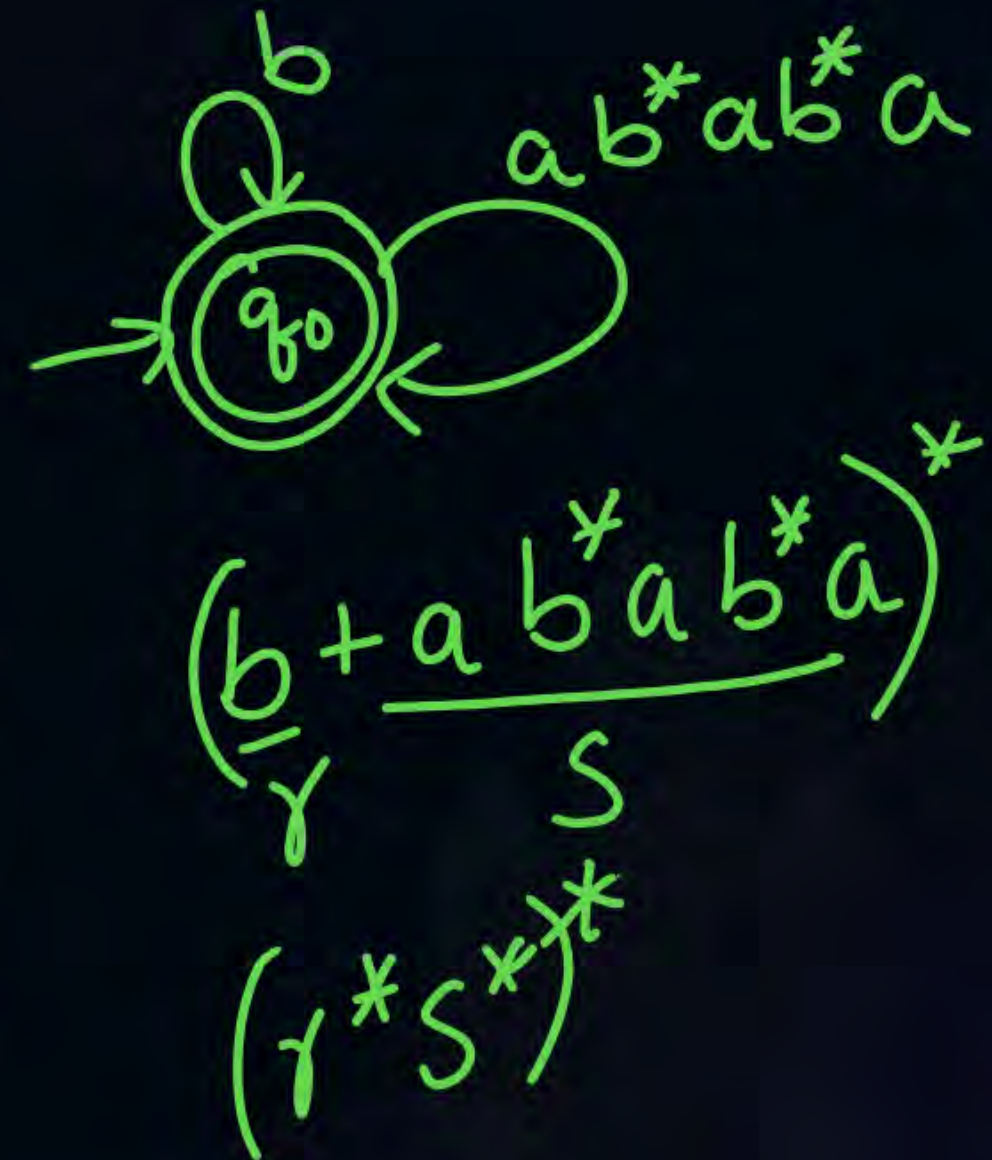
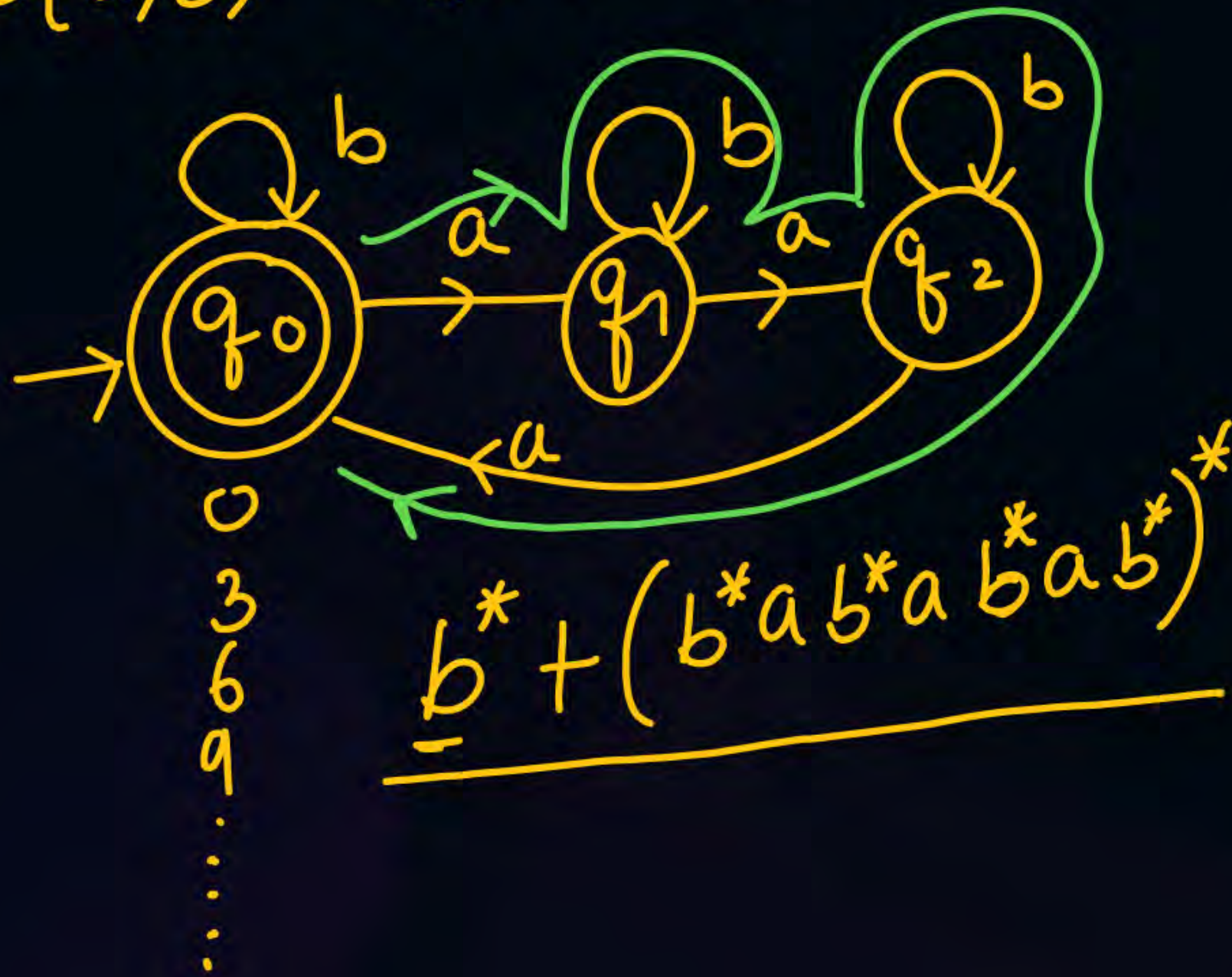
$$L = \{w / n_a(w) \text{ are odd}\}$$

$$\underbrace{(b + ab^*a)^*}_{\text{even no a's}} \underbrace{a}_{\text{odd no a's}} = (b + ab^*a)^* a$$

min DFA



$\Sigma = \{a, b\}$ $n_a(w) \equiv \underline{0} \pmod{\underline{3}}$ ✓



3 minutes

~~min~~ DFA
min NFA



odd no of a's

odd no of a's and even no of b's

Even no of b's



$a - 0 + b - e = \text{odd}$
 $a - e + b - 0 \rightarrow \text{odd}$

odd no of b's

OR $\rightarrow$ union

glcd machine $\rightarrow$ No RE is needed
 $\downarrow$
 generally not asked

even no of a's
 and odd no of b's

Glcd machine

Peter liny →

all man → diff



THANK - YOU